King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Major Exam I Term 261

Original exam date: 28 September 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. A general nonzero solution of the differential equation

$$x^2y' = xy + 3y^2$$

is

(a) $\dfrac{x}{y} + 3\ln|y| = C$

(b) $\dfrac{x}{y} - 3\ln|x| = C$

(c) $\dfrac{x}{y} + 3\ln|x| = C$

(d) $\dfrac{3x}{y} + \ln|x| = C$

(e) $\dfrac{y}{x} + 3\ln|x| = C$

2. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

(a) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1^2 + x_2 = 0$

(b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_4 = 0$

(c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 - 3x_2 + x_4 = 2$

(d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 \le x_2$

(e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 + x_3 = 0$ and $2x_2 - x_4 = 0$

3. Suppose $y = Ce^{mx}$ is a solution of the differential equation

$$y'' + y' - 6y = 0.$$

If $y(0) = 2$ and $\lim_{x\to\infty} y(x) = 0$, then y is

(a) $y = -2e^{-3x}$

(b) $y = -2e^{-2x}$

(c) $y = 2e^{-3x}$

(d) $y = 2e^{3x}$

(e) $y = 2e^{-2x}$

4. The solution of the initial-value problem

$$(x^2+1)y' + 2xy = xe^x, \qquad y(0) = 1$$

is

(a) $y = \dfrac{(x-2)e^x + 2}{x^2+1}$

(b) $y = \dfrac{(x-1)e^x + 1}{x^2+1}$

(c) $y = \dfrac{(x+1)e^x}{x^2+1}$

(d) $y = \dfrac{xe^x + 1}{x^2+1}$

(e) $y = \dfrac{(x-1)e^x + 2}{x^2+1}$

5. If $k(0) = 2$ and the differential equation

$$\left(2y\cos 2x + 5x^4y^2\right)dx + \left(k(x) + 2x^5y\right)dy = 0$$

is exact, then $k(x)$ is

(a) $2 + 2\sin 2x$

(b) $2 + \cos 2x$

(c) $2 + \sin 2x$

(d) $\sin 2x$

(e) $2 - \sin 2x$

6. A hot cup of coffee is placed in a room whose temperature is 25°C. The temperature of the coffee is initially 85°C, and after 10 minutes it is 55°C. Using Newton's law of cooling, the temperature $T(t)$ after t minutes is

(a) $T(t) = 25 + 60\left(\dfrac{1}{3}\right)^{t/10}$

(b) $T(t) = 25 + 60\left(\dfrac{1}{2}\right)^{10t}$

(c) $T(t) = 25 + 60\left(\dfrac{1}{2}\right)^{t/5}$

(d) $T(t) = 25 + 60\left(\dfrac{1}{2}\right)^{t/10}$

(e) $T(t) = 25 + 60\left(\dfrac{2}{3}\right)^{t/10}$

7. The vectors $u = (1, 0, k)$, $v = (0, 1, 1)$, and $w = (4 - k, 1, 4)$ in $\mathbb{R}^3$ are linearly independent when

(a) $k = 1$ or $k = 3$

(b) $k \neq 1, 3$

(c) $k \neq -1, -3$

(d) $k \neq -1, 3$

(e) $k = -1$ or $k = -3$

8. If

$$a\begin{bmatrix}1\\2\end{bmatrix} + b\begin{bmatrix}1\\-1\end{bmatrix} + c\begin{bmatrix}-1\\0\end{bmatrix} + d\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix},$$

and every solution (a, b, c, d) can be written in the form

$$s(1, 0, p, q) + t(0, 1, r, l),$$

then the value of $\dfrac{p + r}{q + l}$ is

(a) 1

(b) -1

(c) -2

(d) 2

(e) 0

9. If a nonconstant solution of the differential equation

$$yy'' = (y')^2 + y'$$

has tangent line $y = 2x + 1$ at $x = 0$, then the solution is given by

(a) $y = e^{2x}$

(b) $y = \dfrac{1}{2} + \dfrac{1}{2}e^{4x}$

(c) $y = \dfrac{1}{3} + \dfrac{2}{3}e^{3x}$

(d) $y = 3 - 2e^{-x}$

(e) $y = -1 + 2e^{x}$

10. The general solution of the differential equation

$$\left(x^3 + \frac{y}{x}\right) dx + \left(y^2 + \ln x\right) dy = 0$$

is

(a) $x^4 + y \ln x + \dfrac{y^3}{3} = C$

(b) $\dfrac{x^4}{4} + x \ln |y| + \dfrac{y^3}{3} = C$

(c) $\dfrac{x^4}{4} + y \ln x + \dfrac{y^2}{2} = C$

(d) $\dfrac{x^4}{4} + y \ln x + y^3 = C$

(e) $3x^4 + 12y \ln x + 4y^3 = C$

11. A particle moves along a straight line with acceleration

$$a(t) = \frac{1}{t^2}, \qquad t > 0.$$

If $s(t)$ denotes the position of the particle, $s(1) = 0$, and $s'(1) = 1$, then $s(t)$ is

(a) $s(t) = 2t - \ln t - 2$

(b) $s(t) = 2t + \dfrac{1}{t} - 3$

(c) $s(t) = t + (\ln t)^2 - 1$

(d) $s(t) = t - (\ln t)^2 - 1$

(e) $s(t) = \ln t$

12. The solution of the initial-value problem

$$\frac{dy}{dx} = \frac{1 + \sin x}{y - 1}, \qquad y(0) = 2,$$

is

(a) $y = 1 + \sqrt{2x - 2\cos x + 3}$

(b) $y = 1 + \sqrt{2x - 4\cos x + 5}$

(c) $y = 1 + \sqrt{2x - \cos x + 2}$

(d) $y = 1 + \sqrt{2x - 2\cos x + 1}$

(e) $y = 1 + \sqrt{2x + 2\cos x - 1}$

13. A one-parameter family of solutions of the differential equation

$$y' = -4x^2 + 4xy - y^2 + 2$$

is given by

(a) $y = 2x - \dfrac{1}{x + C}$

(b) $y = 2x + \dfrac{1}{C - x}$

(c) $y = 2x + \dfrac{1}{x + C}$

(d) $y = x + \dfrac{1}{x + C}$

(e) $y = 2x + \dfrac{1}{2(x + C)}$

14. By an appropriate substitution, the Bernoulli equation

$$x\sqrt[3]{y^2}\,y' + 5\sqrt[3]{y^5} = x^3$$

can be transformed into the linear equation

(a) $3xv' - 25v = 5x^3$

(b) $3xv' + 25v = 5x^3$

(c) $3xv' - 5v = x^3$

(d) $3xv' - 25v = -5x^3$

(e) $3xv' + 5v = x^3$

15. A general nonzero solution of the differential equation

$$2xy' + y^3e^{-2x} = 2xy$$

is given by

(a) $y^2 = e^{-2x}\left(\ln|x| + C\right)$

(b) $y^{-2} = e^{-2x}\left(\dfrac{1}{2}\ln|x| + C\right)$

(c) $y^{-2} = e^{-2x}\left(-\ln|x| + C\right)$

(d) $y^{-2} = e^{2x}\left(\ln|x| + C\right)$

(e) $y^{-2} = e^{-2x}\left(\ln|x| + C\right)$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(c)**	Homogeneous, $v = y/x$: $xv' = 3v^2$, $-\frac{1}{v} = 3\ln\lvert x\rvert + C$, so $\frac{x}{y} + 3\ln\lvert x\rvert = C$.
2	**(e)**	Only the homogeneous linear system $x_1 + x_2 + x_3 = 0,\ 2x_2 - x_4 = 0$ gives a subspace.
3	**(c)**	$m^2 + m - 6 = 0 \Rightarrow m = -3$ (decay), $C = y(0) = 2$: $y = 2e^{-3x}$.
4	**(e)**	$\rho = x^2 + 1$: $((x^2+1)y)' = xe^x$, so $(x^2+1)y = (x-1)e^x + 2$.
5	**(c)**	$M_y = N_x \Rightarrow k' = 2\cos 2x$, so $k = \sin 2x + C$ with $k(0) = 2$: $k = 2 + \sin 2x$.
6	**(d)**	$T = 25 + 60e^{-kt}$, $e^{-10k} = \frac{30}{60} = \frac{1}{2}$: $T = 25 + 60(\frac{1}{2})^{t/10}$.
7	**(b)**	$\det = k^2 - 4k + 3 = (k-1)(k-3) \neq 0$, so $k \neq 1, 3$.
8	**(c)**	$c = a + b$, $d = b - 2a$: $(1,0,1,-2)$, $(0,1,1,1)$, so $\frac{1+1}{-2+1} = -2$.
9	**(c)**	$p = y'$: $y\frac{dp}{dy} = p + 1$, $p + 1 = 3y$, so $y' = 3y - 1$ and $y = \frac{1}{3} + \frac{2}{3}e^{3x}$.
10	**(e)**	Exact ($M_y = N_x = \frac{1}{x}$): $\frac{x^4}{4} + y\ln x + \frac{y^3}{3} = C$, times 12.
11	**(a)**	$v = 2 - \frac{1}{t}$ (from $v(1) = 1$), then $s = 2t - \ln t - 2$ (from $s(1) = 0$).
12	**(a)**	Separable: $\frac{(y-1)^2}{2} = x - \cos x + \frac{3}{2}$, so $y = 1 + \sqrt{2x - 2\cos x + 3}$.
13	**(c)**	$v = 2x - y$: $v' = v^2$, $v = -\frac{1}{x+C}$, so $y = 2x + \frac{1}{x+C}$.
14	**(b)**	$n = -\frac{2}{3}$, $v = y^{5/3}$: $v' + \frac{25}{3x}v = \frac{5}{3}x^2$, i.e. $3xv' + 25v = 5x^3$.
15	**(e)**	Bernoulli $n = 3$, $v = y^{-2}$: $v' + 2v = \frac{e^{-2x}}{x}$, $\rho = e^{2x}$, $y^{-2} = e^{-2x}(\ln\lvert x\rvert + C)$.